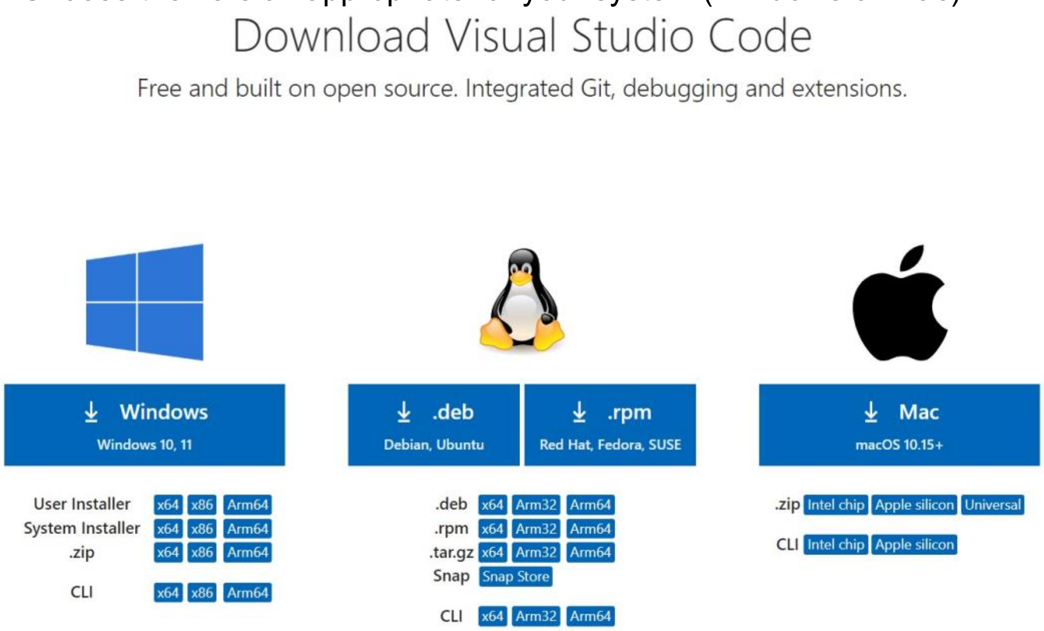
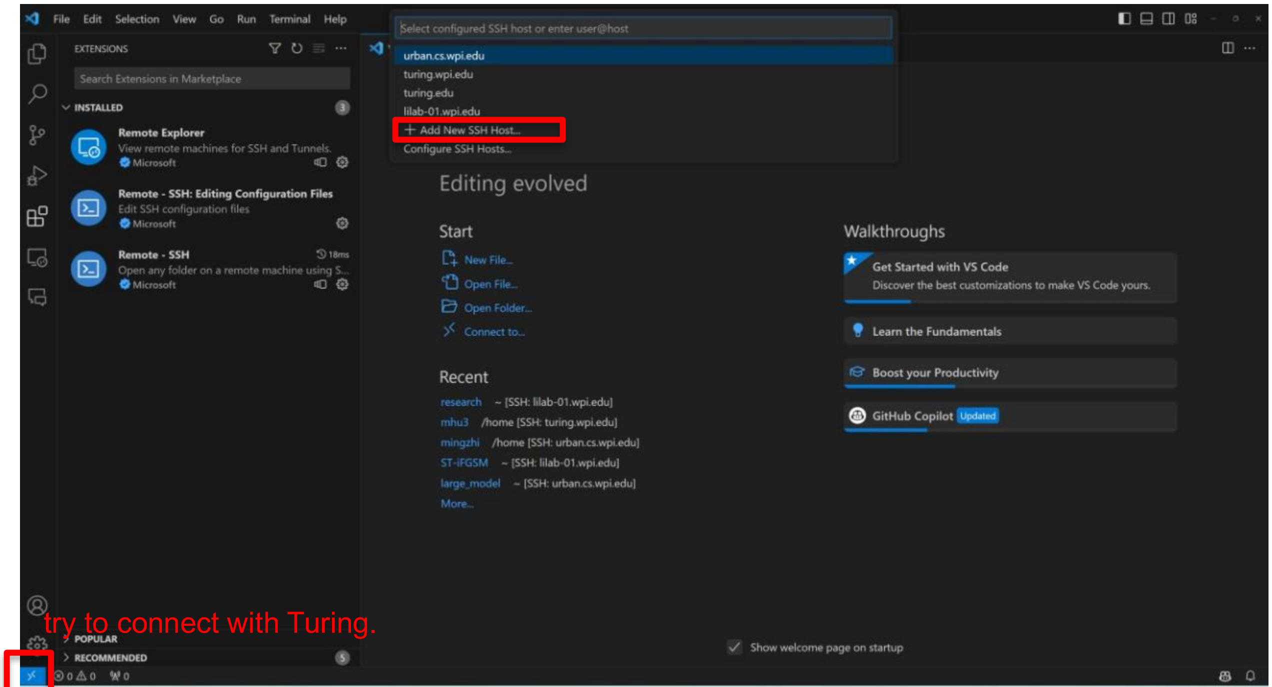
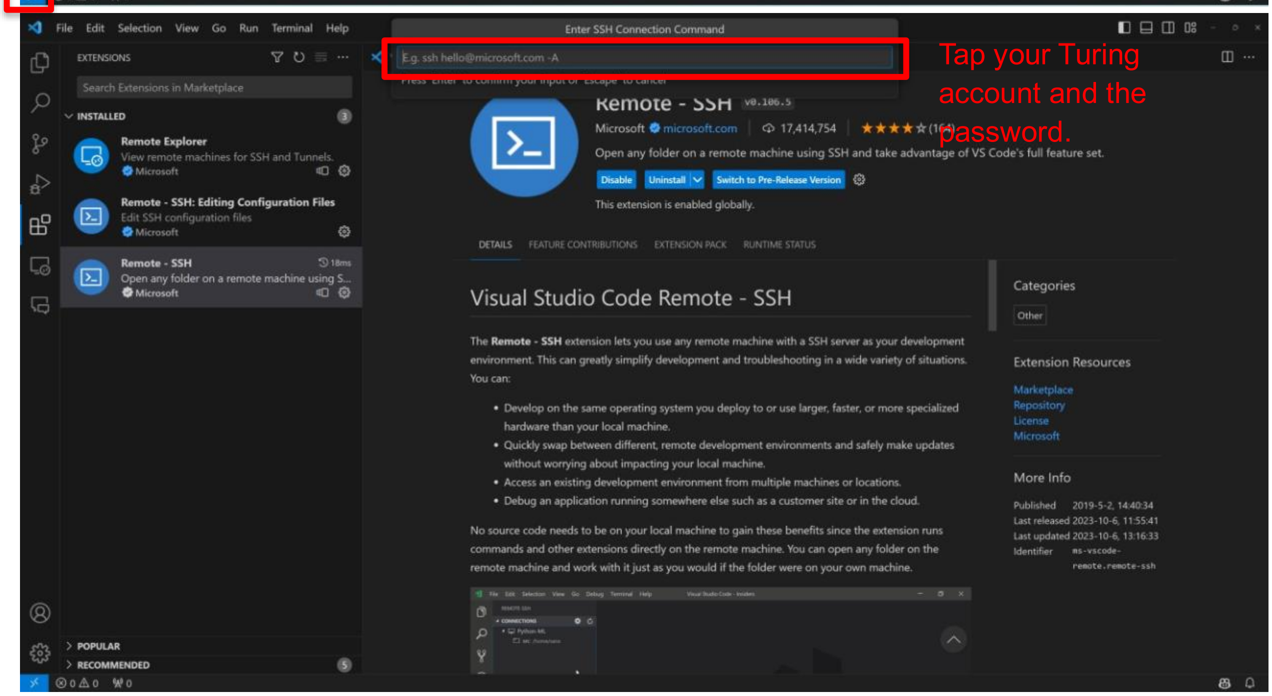
Turing GPU and Environment Tutorial

1. Vscode Installation:
    - Download and setup Vscode on your laptop using this link: https://code.visualstudio.com/download.
    - Choose the version appropriate for your system (Windows or Mac).

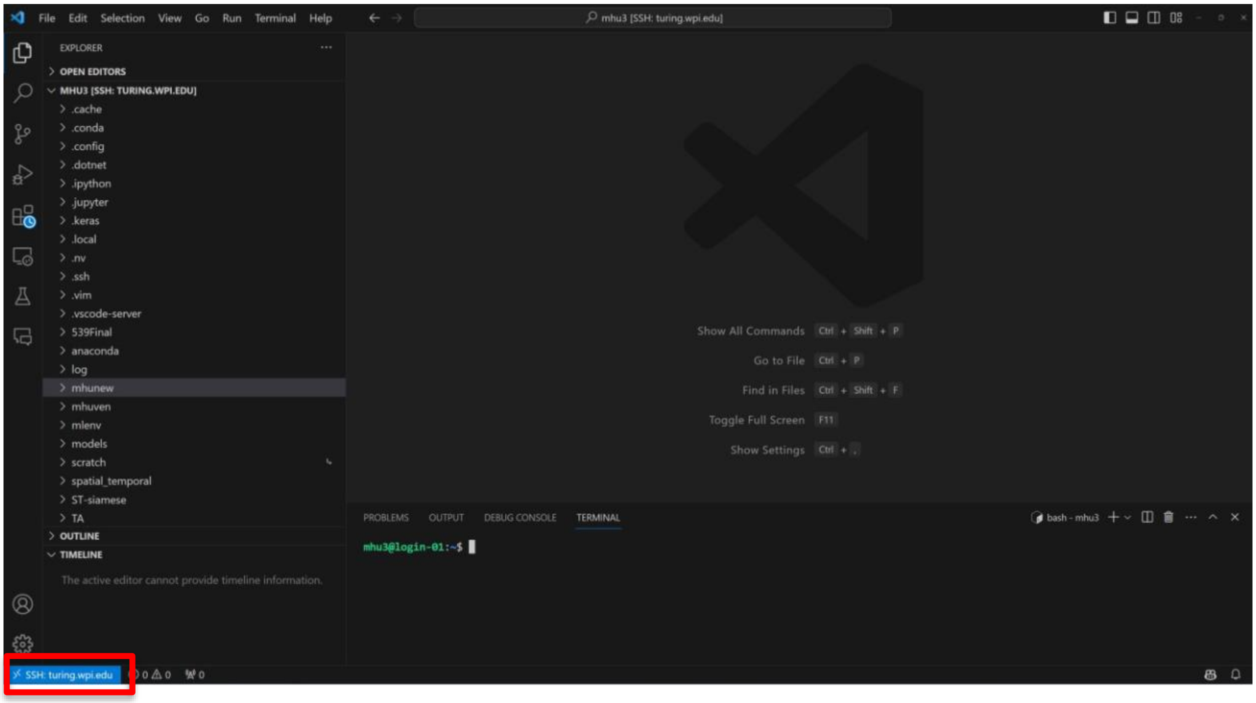


2. Connect to the WPI Network
    - Connect directly if you're on campus.
    - Use the VPN following (https://hub.wpi.edu/article/444/globalprotect-vpnclient-configuration).
3. Open Vscode and try to connect to the Turing:
   Vscode will automatically install the extensions of SSH for you.



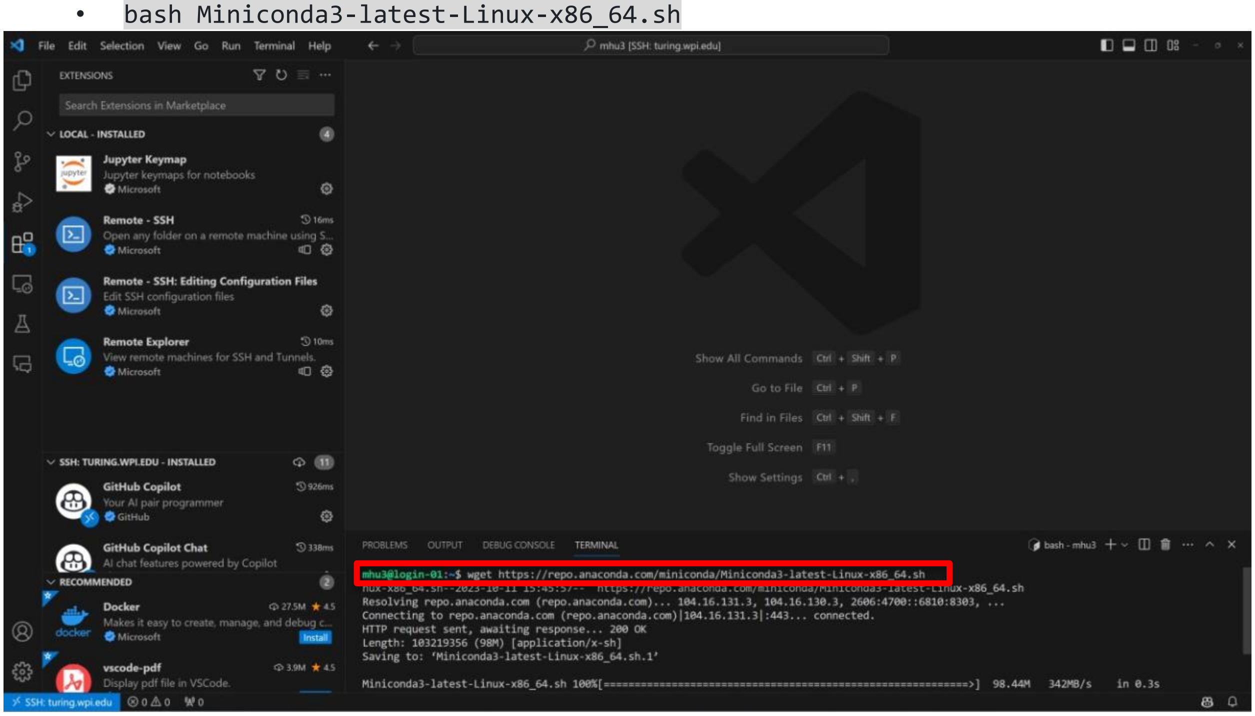


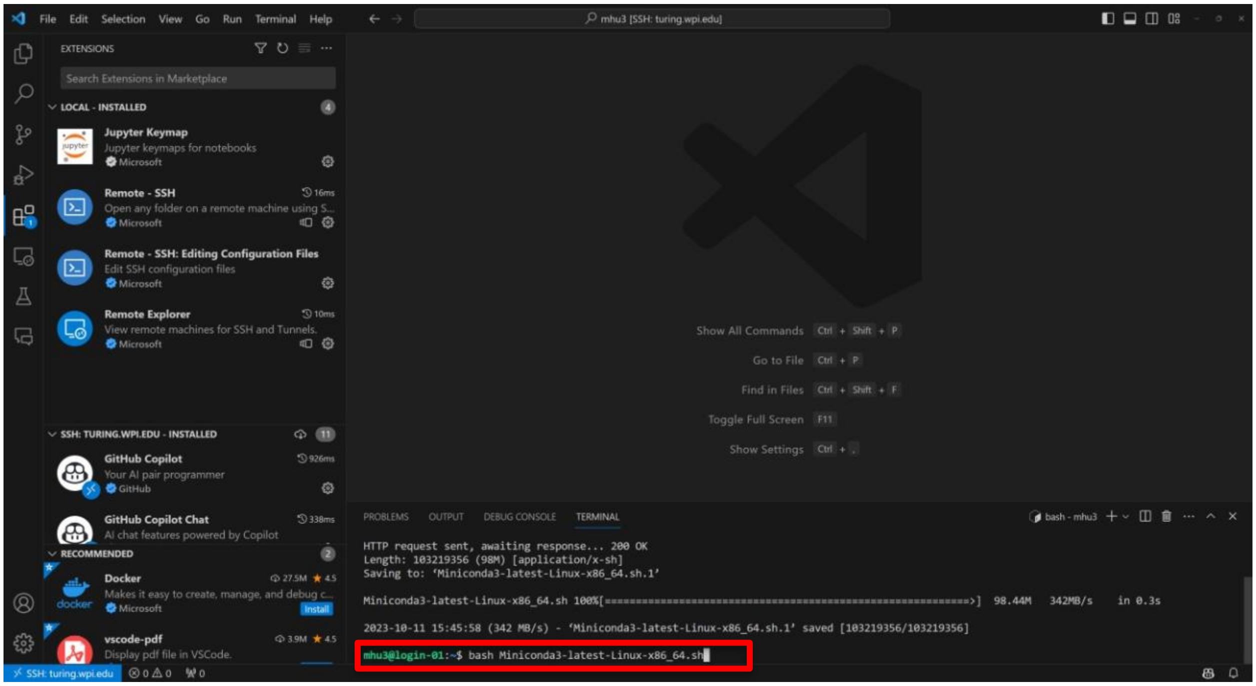
Now you are connected to Turing.



4. Download conda to the server and install it.

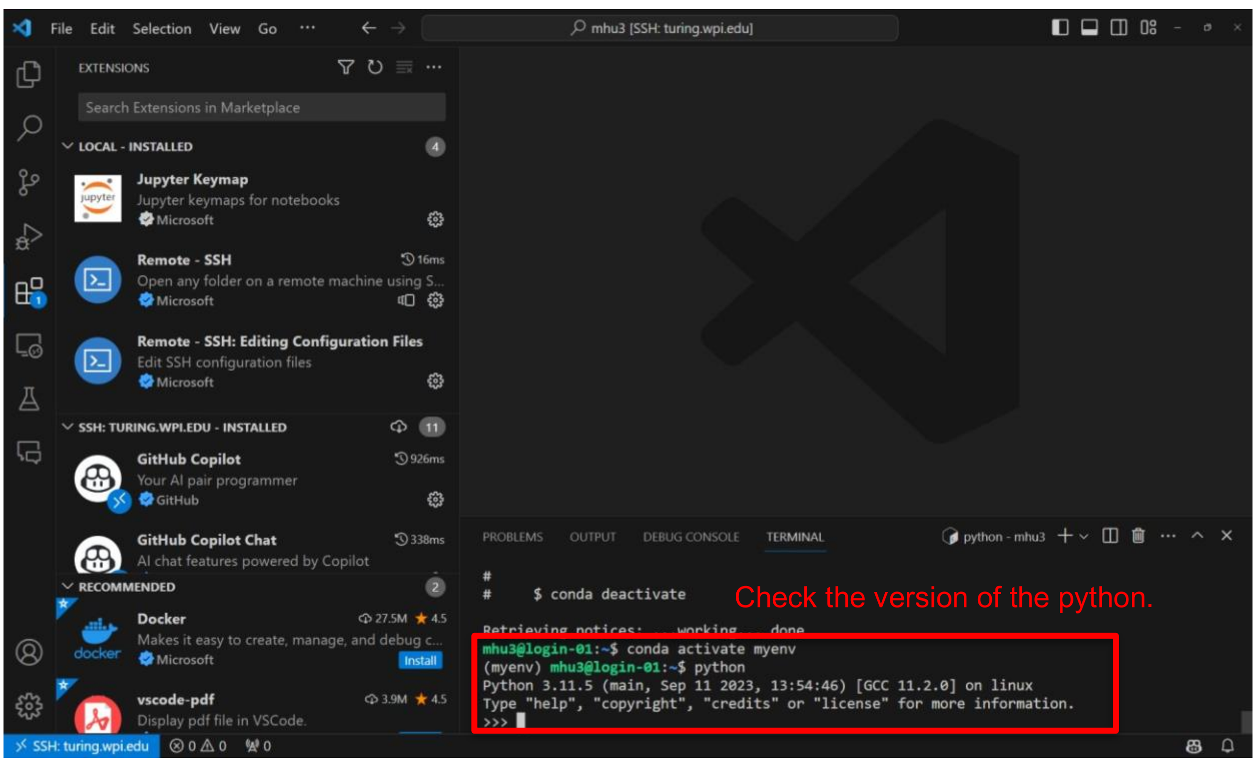
- `wget https://repo.anaconda.com/miniconda/Miniconda3-latest-Linux-x86_64.sh`
- `bash Miniconda3-latest-Linux-x86_64.sh`



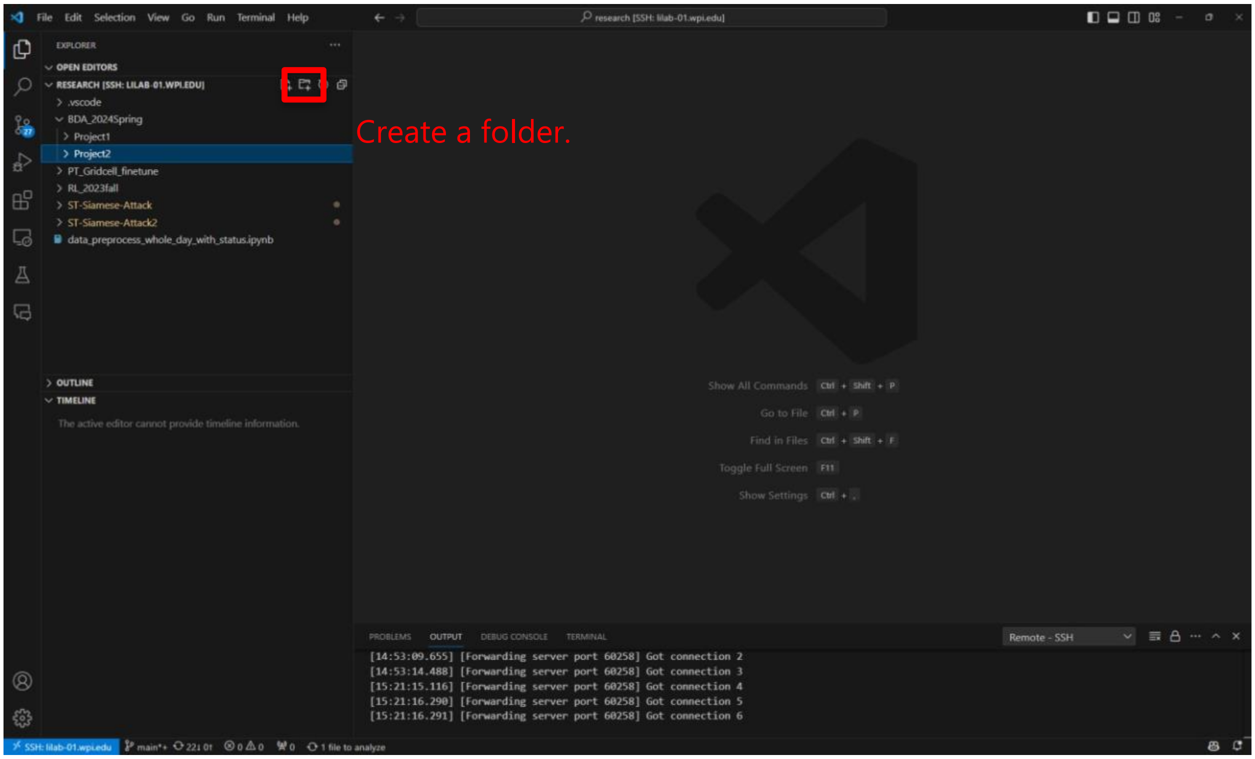


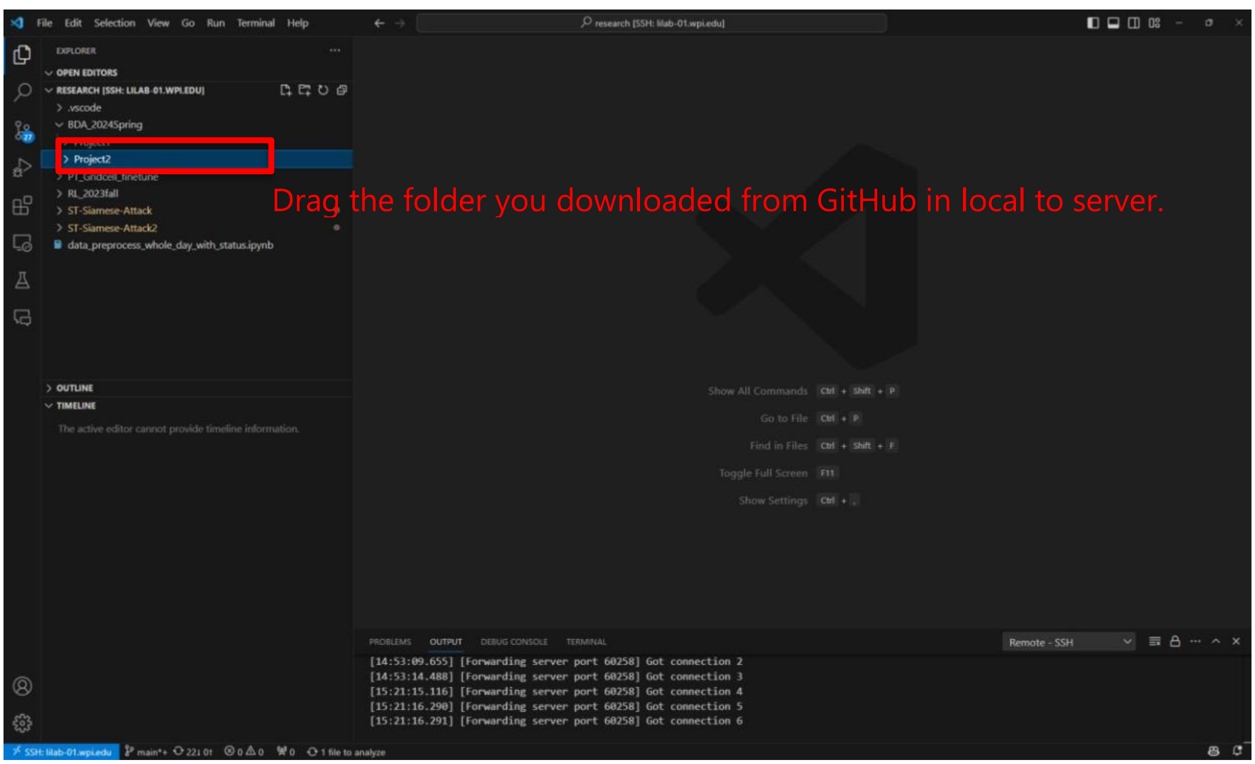
After the installation is done, restart your bash by typing bash if needed. conda will create an environment named "base" for you.

5. Create virtual environment: `conda create -n myenv python==3.11.5.`
6. After creating the virtual environment, activate it: `conda activate myenv` please install the required packages in the environment.

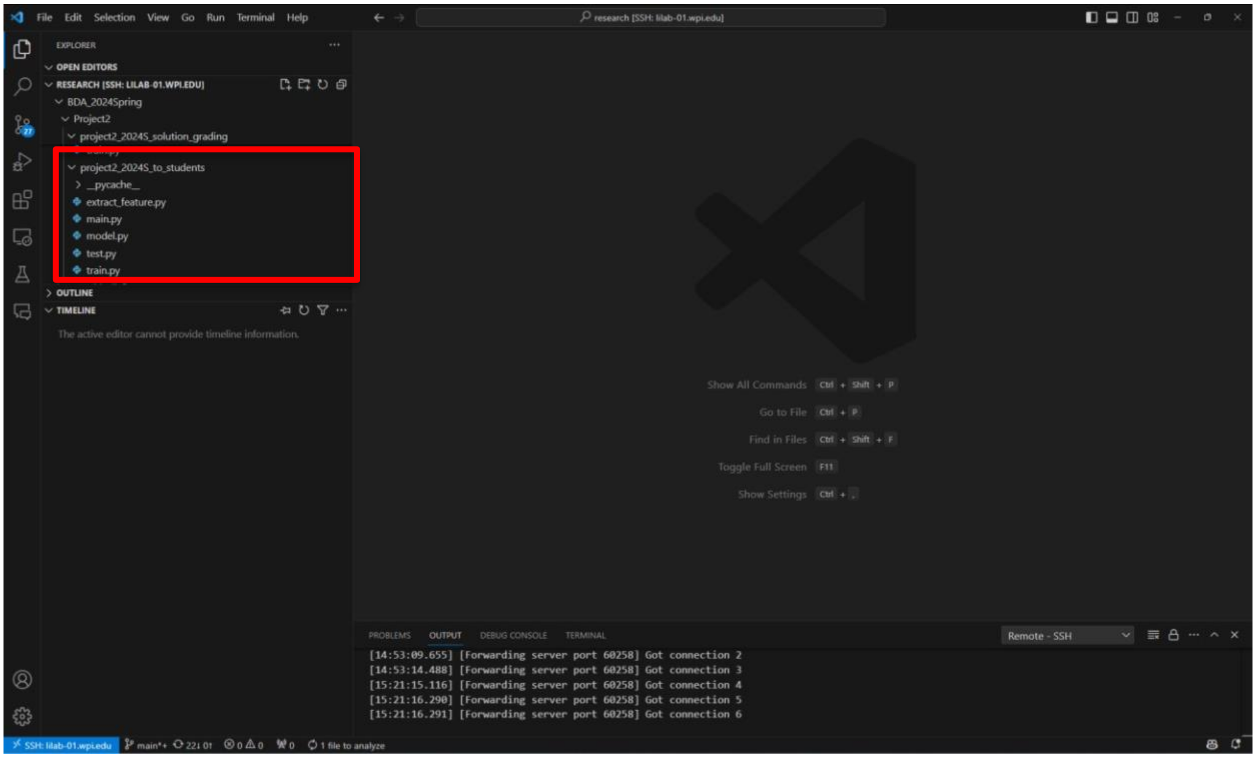


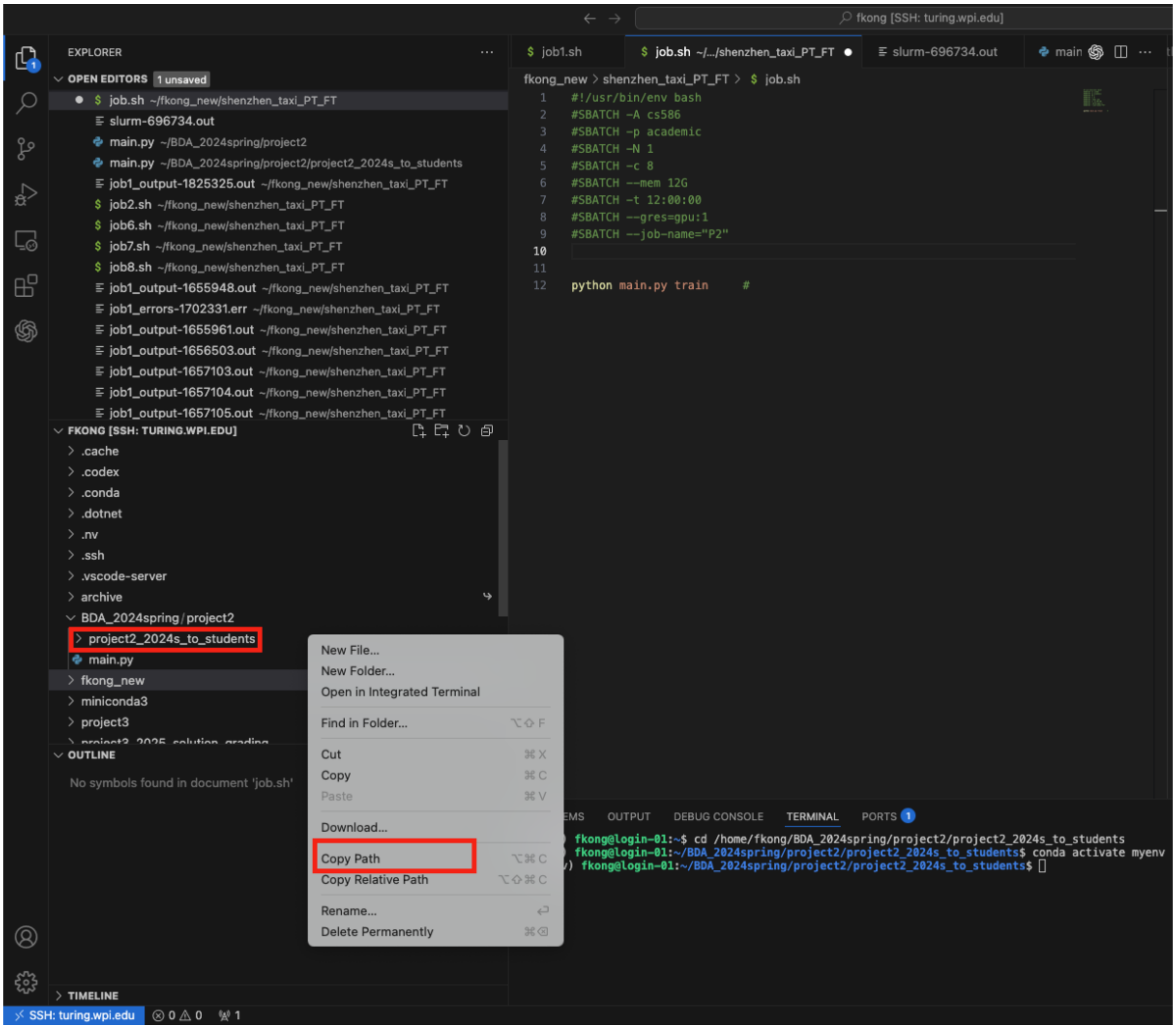
7. Project Directory Setup. Create a folder for the project.



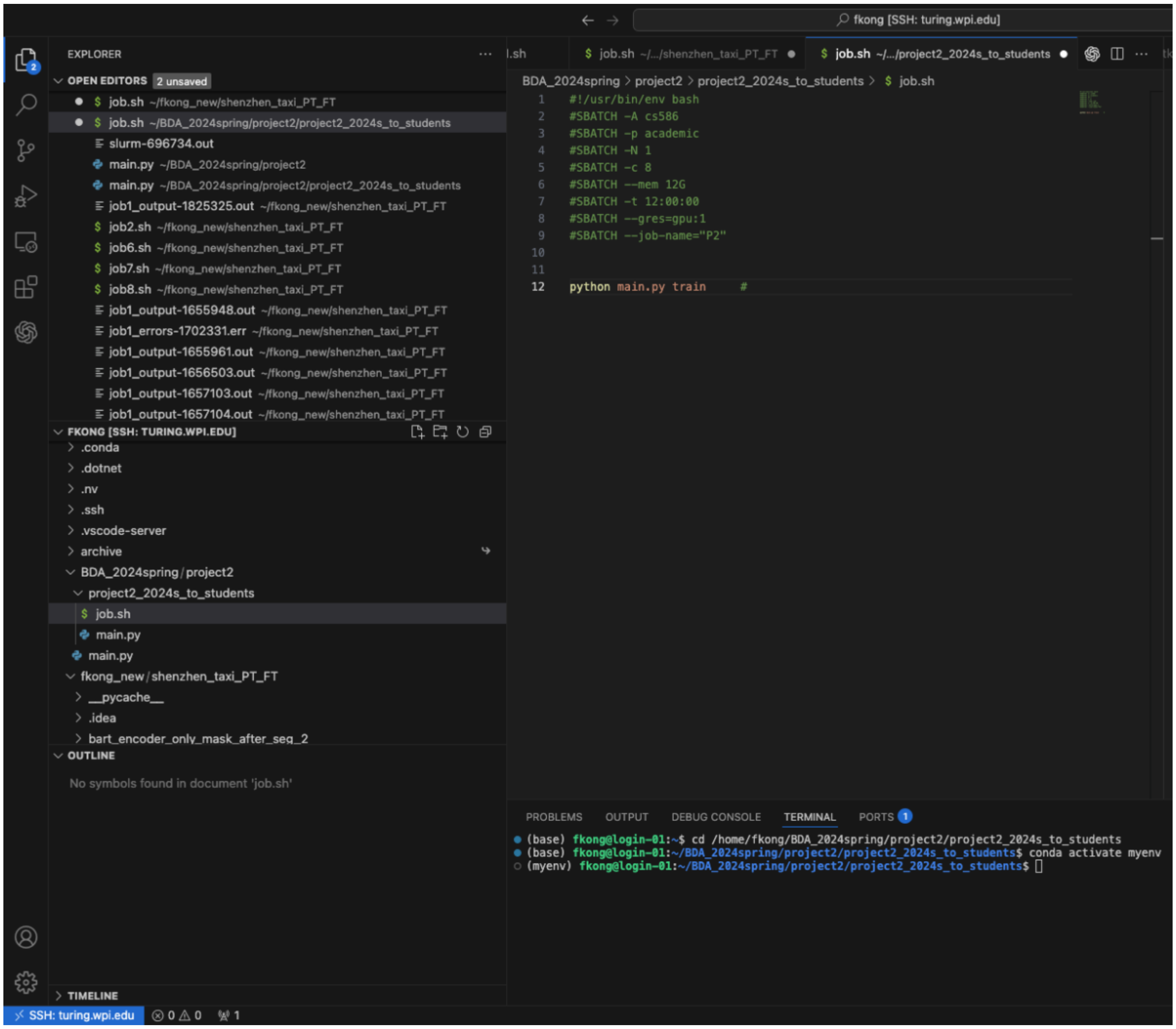


8. Navigate to the Directory using `cd path` ("path" is the dictionary you have copied.)


File Edit Selection View Go Run Terminal Help
research [SSH: lilab-01.wpi.edu]
EXPLORER
OPEN EDITORS
RESEARCH [SSH: LILAB-01.WPI.EDU]
BDA_2024Spring
Project2
project2_2024S_solution_grading
project2_2024S_to_students
__pycache__
extract_feature.py
main.py
model.py
test.py
train.py
OUTLINE
TIMELINE
The active editor cannot provide timeline information.
Show All Commands Ctrl + Shift + P
Go to File Ctrl + P
Find in Files Ctrl + Shift + F
Toggle Full Screen F11
Show Settings Ctrl + ,
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL
Remote - SSH
[14:53:09.655] [Forwarding server port 60258] Got connection 2
[14:53:14.488] [Forwarding server port 60258] Got connection 3
[15:21:15.116] [Forwarding server port 60258] Got connection 4
[15:21:16.290] [Forwarding server port 60258] Got connection 5
[15:21:16.291] [Forwarding server port 60258] Got connection 6
SSH: lilab-01.wpi.edu
1 file to analyze

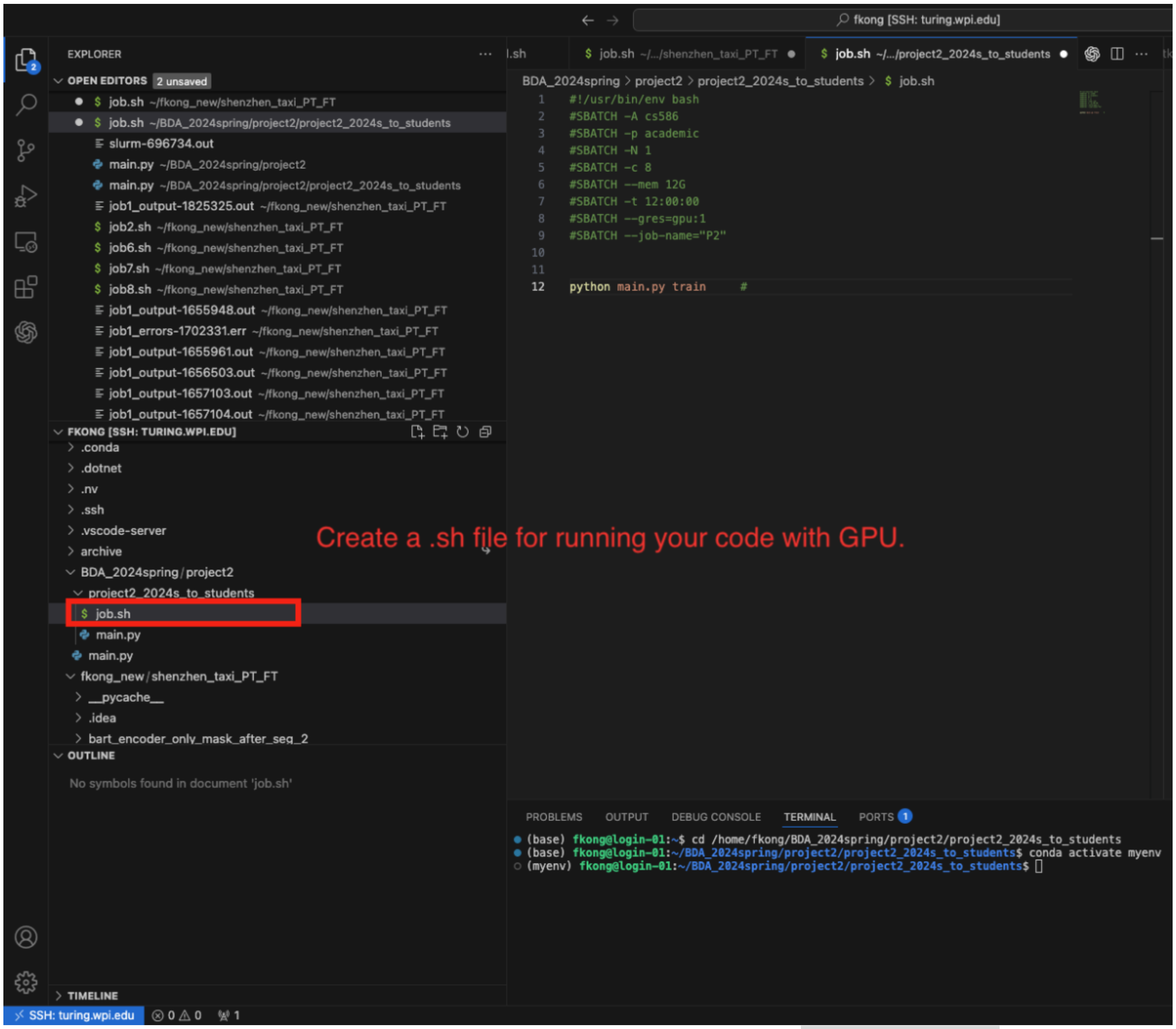


fkong [SSH: turing.wpi.edu]
EXPLORER
OPEN EDITORS 1 unsaved
job.sh ~/fkong_new/shenzhen_taxi_PT_FT
slurm-696734.out
main.py ~/BDA_2024spring/project2
main.py ~/BDA_2024spring/project2/project2_2024s_to_students
job1_output-1825325.out ~/fkong_new/shenzhen_taxi_PT_FT
job2.sh ~/fkong_new/shenzhen_taxi_PT_FT
job6.sh ~/fkong_new/shenzhen_taxi_PT_FT
job7.sh ~/fkong_new/shenzhen_taxi_PT_FT
job8.sh ~/fkong_new/shenzhen_taxi_PT_FT
job1_output-1655948.out ~/fkong_new/shenzhen_taxi_PT_FT
job1_errors-1702331.err ~/fkong_new/shenzhen_taxi_PT_FT
job1_output-1655961.out ~/fkong_new/shenzhen_taxi_PT_FT
job1_output-1656503.out ~/fkong_new/shenzhen_taxi_PT_FT
job1_output-1657103.out ~/fkong_new/shenzhen_taxi_PT_FT
job1_output-1657104.out ~/fkong_new/shenzhen_taxi_PT_FT
job1_output-1657105.out ~/fkong_new/shenzhen_taxi_PT_FT
FKONG [SSH: TURING.WPI.EDU]
.cache
.codex
.conda
.dotnet
.nv
.ssh
.vscode-server
archive
BDA_2024spring / project2
project2_2024s_to_students
main.py
fkong_new
miniconda3
project3
OUTLINE
No symbols found in document 'job.sh'
TIMELINE
job1.sh
job.sh ~/.../shenzhen_taxi_PT_FT
slurm-696734.out
main
fkong_new > shenzhen_taxi_PT_FT > job.sh
#!/usr/bin/env bash
#SBATCH -A cs586
#SBATCH -p academic
#SBATCH -N 1
#SBATCH -c 8
#SBATCH --mem 12G
#SBATCH -t 12:00:00
#SBATCH --gres=gpu:1
#SBATCH --job-name="P2"
python main.py train #
New File...
New Folder...
Open in Integrated Terminal
Find in Folder...
Cut
Copy
Paste
Download...
Copy Path
Copy Relative Path
Rename...
Delete Permanently
OUTPUT
DEBUG CONSOLE
TERMINAL
PORTS 1
fkong@login-01:~$ cd /home/fkong/BDA_2024spring/project2/project2_2024s_to_students
fkong@login-01:~/BDA_2024spring/project2/project2_2024s_to_students$ conda activate myenv
fkong@login-01:~/BDA_2024spring/project2/project2_2024s_to_students$
SSH: turing.wpi.edu



9. Prepare Your Job:
Ensure you have a script (e.g., job.sh) ready. This script should specify the necessary resources (like CPU cores, RAM, GPU) and contain the instructions for running your job. Please make sure you have the following lines in the script.

```
#SBATCH -A cs586
#SBATCH -p academic
```



10. Use the following command to submit your job: `sbatch job.sh`.
11. Launch an Interactive Session for Testing: Use `sinteractive` .

This command is particularly useful when you want to test your code on the server's resources before running the actual job. Note that in an interactive session:

- Your terminal session must remain active for the duration of the session.
- Any code you run will execute immediately, unlike the batch mode (sbatch), which queues your job to be run based on the server's schedule and available resources.

After running this command, wait for the server (in this case, Turing) to allocate resources to you, such as a GPU. Once resources are assigned and your prompt changes, you're in an interactive session and can run commands as if you're on the allocated node.